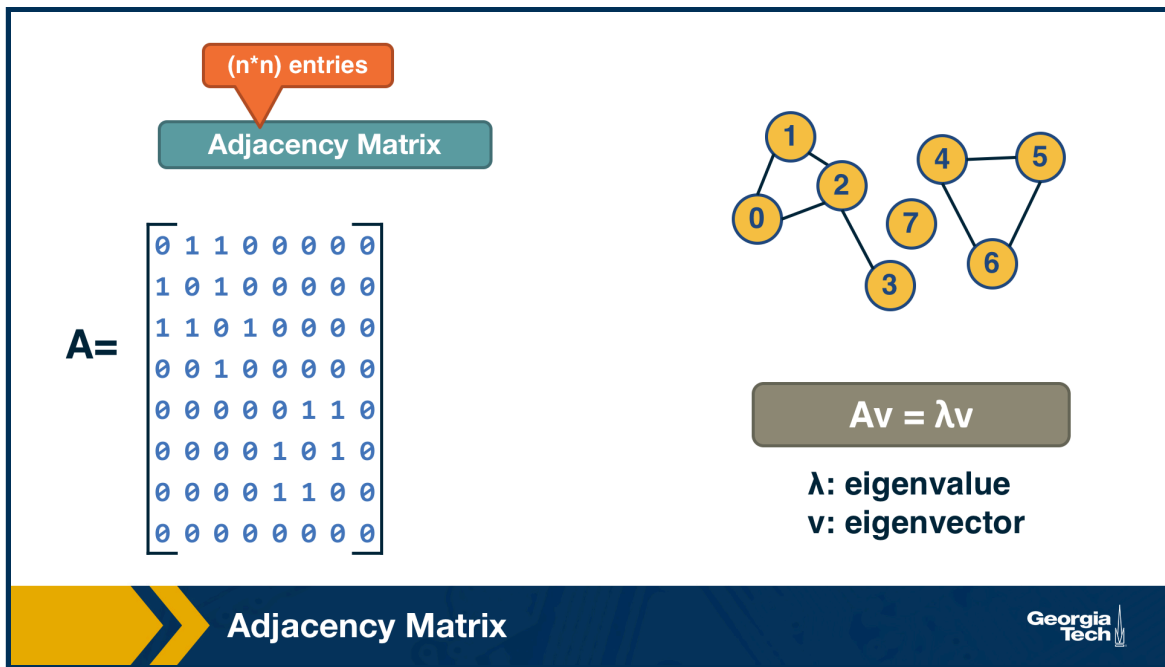


L2: Adjacency Matrix



A graph is often represented either with an **Adjacency Matrix**, as shown in this visualization. The matrix representation requires a single memory access to check if an edge exists but it requires n^2 space. The adjacency matrix representation allows us to use tools from linear algebra to study graph properties.

For example, an undirected graph is represented by a symmetric matrix A – and so the eigenvalues of A are a set of real numbers (*referred to as the “spectrum” of the graph*). The equation at the right of the visualization reminds you the definition of eigenvalues and eigenvectors.

Food for Thought

How would you show mathematically that the largest eigenvalue of the (*symmetric*) adjacency matrix A is less or equal than the maximum node degree in the network? Start from the definition of eigenvalues given above.